PCBC H Name _____ _____ /22

Limits Quiz 1.1–1.2

Non-Calculator

You have 20 minutes.

PRACTICE B · built to match the real Quiz 1.1–1.2: 11 questions, 20 minutes, no calculator

Honor Code (copy below & sign): On my honor, before God, I have neither given, nor received any aid on this quiz.

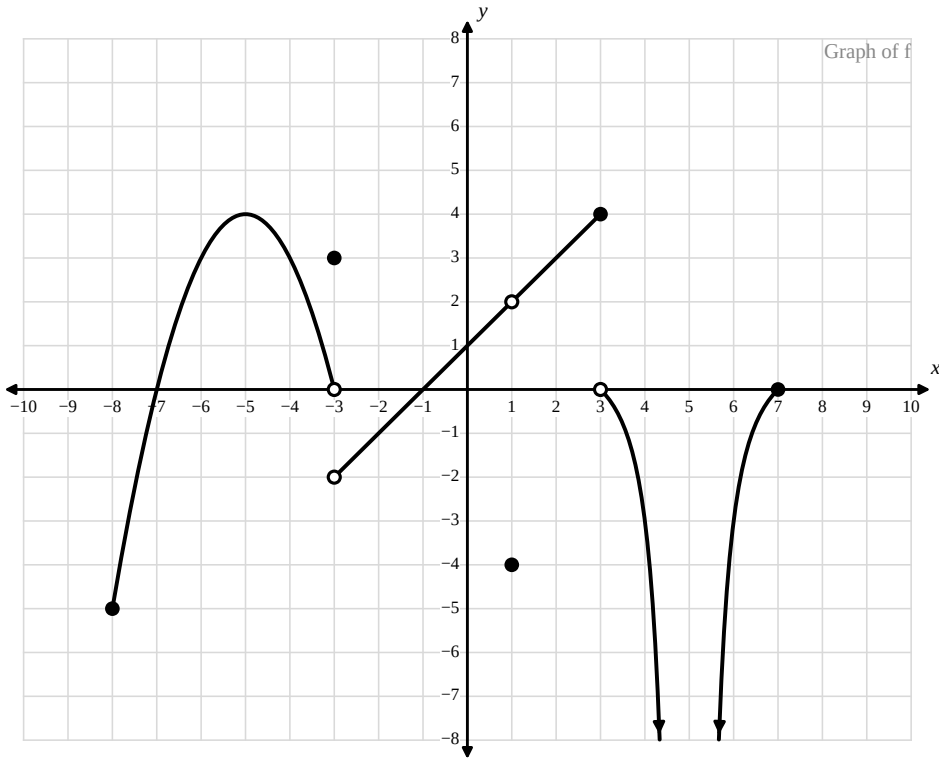
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Signature _____

SHOW ALL WORK FOR FULL CREDIT.

Use the graph of f shown below to find each. If a limit does not exist, write DNE. (2 points each)

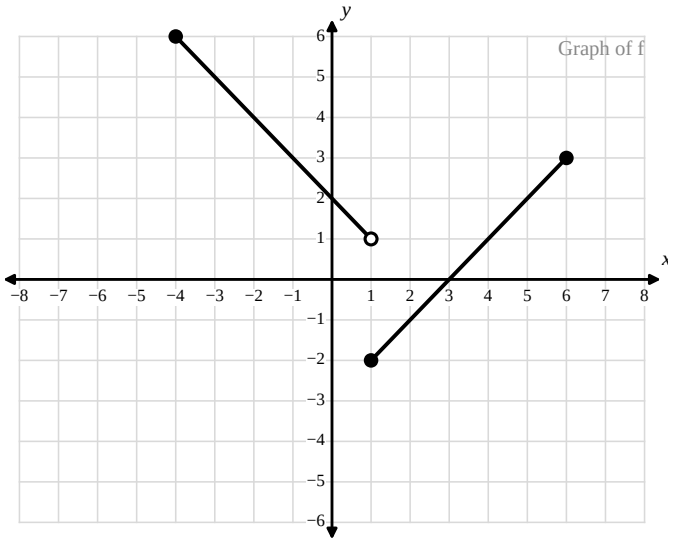


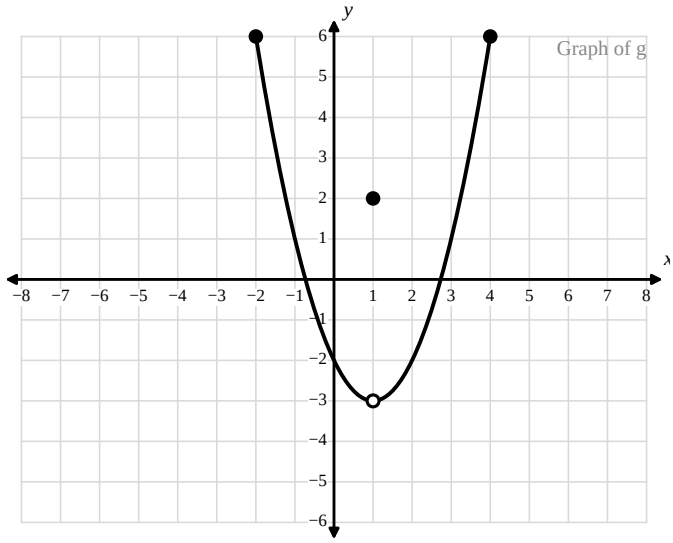
1. $\lim_{x\to 1} f(x) =$	**2.** $f(1) =$	**3.** $\lim_{x\to -3} f(x) =$

(2 points)

4. Use the graph of f above to determine all values of a for which $\lim_{x\to a} f(x) = 0$ on the interval $-9 < x < 9$. If there are no values of a, write the empty set.

Use the graphs of f and g below to find the limit. Show the limit properties you use. (2 points)





5. $\lim_{x \to 1} [f(x) + g(x)]$

(2 points)

6. Use the table to estimate $\lim_{x \to -1} f(x)$. Give the left-hand limit, the right-hand limit, and the limit, or state that it does not exist.

x	−1.1	−1.01	−1.001	$\to -1 \leftarrow$	−0.999	−0.99	−0.9
$f(x)$	3.21	3.0201	3.002001		−0.998	−0.98	−0.8

$\lim_{x \to -1^-} f(x) \approx$ _____ $\lim_{x \to -1^+} f(x) \approx$ _____ $\lim_{x \to -1} f(x) \approx$ _____

Evaluate each limit. (2 points each)

7. $\lim\limits_{x \to 3} \dfrac{x+2}{x-3}$

8. $\lim\limits_{x \to 2} \dfrac{x^3 - 8}{x^2 - 4}$

9. $\lim\limits_{x \to 6} \dfrac{\sqrt{x+3} - 3}{x - 6}$

10. $\lim\limits_{x \to -1} \dfrac{\frac{1}{x+3} - \frac{1}{2}}{x+1}$

11. For $f(x) = \sqrt{x-2}$, find $\lim\limits_{h \to 0} \dfrac{f(x+h) - f(x)}{h}$

Each step: the work on the left, the reason on the right. "lim" stays on every line until the line where you substitute.

1. $\boxed{2}$

Both sides of the line head to the open circle at $(1, 2)$.	Ignore the filled dot at $(1, -4)$.

2. $\boxed{-4}$

The filled dot at $x = 1$ is at height -4.	Function value.

3. $\boxed{\text{DNE}}$

$\lim\limits_{x \to -3^-} f(x) = 0, \lim\limits_{x \to -3^+} f(x) = -2$	Left: the parabola ends at the open circle $(-3, 0)$. Right: the line starts at the open circle $(-3, -2)$.
$0 \neq -2$, so $\lim_{x \to -3} f(x)$ does not exist.	The dot $(-3, 3)$ is $f(-3)$ and does not matter.

4. $\boxed{a = -7,\ a = -1}$

Height 0 is reached from both sides at $x = -7$ and $x = -1$.	Two-sided approach to $y = 0$.
Not $x = -3$ (left side only), not $x = 3$ (right side only; the left heads to 4), not $x = 7$ (graph ends there).	One-sided approaches and endpoints do not count.

5. $\boxed{\text{DNE}}$

$\lim\limits_{x \to 1^-} [f(x) + g(x)] = 1 + (-3) = -2$	Left: $f \to 1$, $g \to -3$.
$\lim\limits_{x \to 1^+} [f(x) + g(x)] = -2 + (-3) = -5$	Right: $f \to -2$ (filled dot), $g \to -3$.
$-2 \neq -5$, so $\lim_{x \to 1} [f(x) + g(x)]$ does not exist.	You cannot split the two-sided limit because $\lim_{x \to 1} f(x)$ does not exist; work each side.

6. $\boxed{\text{DNE (suggested)}}$

Left: 3.21, 3.0201, 3.002001	Heading to 3.
Right: $-0.8,\ -0.98,\ -0.998$	Heading to -1.
$\lim\limits_{x \to -1^-} f(x) \approx 3, \quad \lim\limits_{x \to -1^+} f(x) \approx -1$	The two trends approach different values, so the table suggests $\lim_{x \to -1} f(x)$ does not exist. A table is evidence, not proof.

7. $\boxed{\text{DNE}}$

Substitution produces $\frac{5}{0}$: a nonzero number over 0.	No finite limit. The limit does not exist; done.
Left: top near 5, bottom tiny negative, so $-\infty$. Right: $+\infty$.	Opposite directions, so $\lim_{x \to 3} \frac{x+2}{x-3}$ does not exist.

8. $\boxed{3}$

Substitution produces 0/0: indeterminate.	Factor.
$\lim\limits_{x \to 2} \dfrac{x^3 - 8}{x^2 - 4} = \lim\limits_{x \to 2} \dfrac{(x - 2)\left(x^2 + 2x + 4\right)}{(x - 2)(x + 2)}$	Difference of cubes; difference of squares.
$= \lim\limits_{x \to 2} \dfrac{x^2 + 2x + 4}{x + 2}$	Cancel $x - 2$.
$= \dfrac{4 + 4 + 4}{2 + 2} = \dfrac{12}{4} = 3$	Substitute.

9. $\boxed{\frac{1}{6}}$

Substitution produces 0/0: indeterminate.	Multiply by the conjugate $\sqrt{x+3}+3$.
$\lim\limits_{x\to 6}\dfrac{\sqrt{x+3}-3}{x-6}=\lim\limits_{x\to 6}\dfrac{(x+3)-9}{(x-6)\left(\sqrt{x+3}+3\right)}$	Top is a^2-b^2. Do not FOIL; leave the bottom factored.
$=\lim\limits_{x\to 6}\dfrac{x-6}{(x-6)\left(\sqrt{x+3}+3\right)}=\lim\limits_{x\to 6}\dfrac{1}{\sqrt{x+3}+3}$	Cancel $x-6$.
$=\dfrac{1}{\sqrt{9}+3}=\dfrac{1}{6}$	Substitute.

10. $\boxed{-\frac{1}{4}}$

Substitution produces 0/0: indeterminate.	Multiply top and bottom by the LCD $2(x+3)$.
$\lim\limits_{x\to -1}\dfrac{\frac{1}{x+3}-\frac{1}{2}}{x+1}=\lim\limits_{x\to -1}\dfrac{2-(x+3)}{2(x+3)(x+1)}$	$2(x+3)\cdot\frac{1}{x+3}=2$, $2(x+3)\cdot\frac{1}{2}=x+3$.
$=\lim\limits_{x\to -1}\dfrac{-(x+1)}{2(x+3)(x+1)}$	$2-x-3=-x-1=-(x+1)$.
$=\lim\limits_{x\to -1}\dfrac{-1}{2(x+3)}=\dfrac{-1}{2(2)}=-\dfrac{1}{4}$	Cancel $x+1$; substitute.

11. $\boxed{\frac{1}{2\sqrt{x-2}}}$

$\lim\limits_{h\to 0}\dfrac{\sqrt{x+h-2}-\sqrt{x-2}}{h}$	$f(x+h)=\sqrt{(x+h)-2}$. Substituting $h=0$ produces 0/0.
$=\lim\limits_{h\to 0}\dfrac{(x+h-2)-(x-2)}{h\left(\sqrt{x+h-2}+\sqrt{x-2}\right)}$	Multiply by the conjugate.
$=\lim\limits_{h\to 0}\dfrac{h}{h\left(\sqrt{x+h-2}+\sqrt{x-2}\right)}=\lim\limits_{h\to 0}\dfrac{1}{\sqrt{x+h-2}+\sqrt{x-2}}$	Cancel h.
$=\dfrac{1}{2\sqrt{x-2}}$	Substitute 0 for h. Needs $x>2$.